

## ALGORITHMIC SOLUTIONS FOR FAIR CREDIT SCORING

A Abhiram<sup>\*1</sup>, B Srinivas<sup>\*2</sup>, Ch Sushruth<sup>\*3</sup>, Ch Yasaswi<sup>\*4</sup>, G. Joel Krupkar<sup>\*5</sup>

<sup>\*1,2,3,4</sup>Student, Department Of IT. Malla Reddy Engineering College, Maisammaguda, Hyderabad, India.

<sup>\*5</sup>Professor, Department Of IT. Malla Reddy Engineering College, Maisammaguda, Hyderabad, India.

### ABSTRACT

The success of machine learning in assessing the creditworthiness of borrowers has been recognized for quite some time. Nevertheless, there are apprehensions that the implementation of automated decision-making systems might lead to unequal treatment of certain groups or individuals, which could result in biased outcomes. This study aims to tackle this concern by analyzing the effectiveness of 12 prominent bias reduction strategies across 5 different fairness indicators, while also evaluating their accuracy and potential financial benefits for lending institutions. Our investigation has revealed the difficulties in achieving fairness alongside accuracy and profitability, and we have pointed out both the most effective and least effective bias mitigation approaches. Ultimately, our research intends to connect the divide between experimental machine learning and its real-world applications within the financial sector.

**Keywords:** Bias Mitigation, Credit Scoring, Algorithmic Decision, Fair AI.

### I. INTRODUCTION

Credit scoring applications have become a crucial component of modern financial systems, with loan approvals increasingly transitioning from human decision-making to algorithmic processes. Agarwal et al. (2020) highlighted several benefits of automated decision systems for financial institutions, including enhanced business growth, cost reduction, increased approval rates without escalating credit risks, and a streamlined application process for clients. However, this shift raises significant concerns about the supervision of these automated decisions. Ensuring that algorithmic decisions remain transparent and accountable is vital for maintaining fair financial practices. Blattner et al. (2019) emphasized the inherent trade-off between algorithmic complexity and regulatory oversight. While more complex models often deliver better performance, they tend to be less interpretable, complicating efforts to audit and supervise their decisions. Regulators, including national banks and financial authorities, are particularly concerned about the transparency of credit scoring methods. The ability of humans to understand and interpret algorithmic decisions is essential for mitigating risks and preventing financial misconduct. Furthermore, even transparent automated decisions may result in discriminatory practices, disproportionately affecting certain groups based on specific attributes (Kleinberg et al., 2018). Although legal frameworks prohibit explicit discrimination based on factors such as gender, race, and nationality, other less regulated information can also be used in harmful ways.

For instance, behavioral and financial data might inadvertently lead to biased outcomes. Additionally, external data sources, including social media, can amplify such biases by reflecting social inequalities (Barocas & Selbst, 2016). Addressing these challenges requires the development of robust mechanisms to ensure algorithmic fairness in credit scoring systems. Concerns regarding the use of artificial intelligence (AI) in decision-making have gained attention from both researchers and policymakers. Recent reports (AI Now Institute, 2021; European Commission, 2020) propose regulatory guidelines emphasizing interpretability, fairness, privacy, technical robustness, and accountability. While the General Data Protection Regulation (GDPR) in the European Union addresses data privacy concerns, additional financial regulations like Basel III and IFRS 9 establish guidelines for transparency and risk management in financial institutions. Effective implementation of these regulations requires the adoption of fair and transparent AI models in credit scoring systems. This study aims to address fairness in automated credit scoring using machine learning algorithms. Recognizing the gap between experimental advancements and real-world applications, the study benchmarks twelve bias mitigation methods using a real-world dataset from the Romanian consumer loan market. The study contributes by evaluating bias mitigation methods based on fairness metrics, model accuracy, and financial institution profitability. Additionally, a new publicly available dataset enables further research in credit scoring fairness. A comprehensive review of fairness metrics and bias mitigation techniques is also presented. The subsequent

sections cover a literature review on fairness in machine learning, detailed methodology on applying fairness processors, experimental setup and results, and concluding insights for future research.

## II. LITERATURE SURVEY

Recent research in the field of machine learning for credit scoring has increasingly focused on the distributional impacts of algorithmic decisions, especially regarding vulnerable or sensitive groups. While traditional methods such as human application analysis have been overtaken by machine learning techniques in terms of profitability for financial institutions, machine learning has shown a greater variance in predictions, which raises concerns about its fairness and accuracy [21]. Zarsky [22] proposed an analytic framework to guide discussions on the ethics of algorithmic decision-making in credit scoring, identifying two primary issues: the efficiency of automated decision-making and the fairness of the process. One notable concern is the automation of the credit scoring process, which, despite yielding strong aggregated results, remains prone to errors due to inaccurate training data or flawed algorithm settings. While human decision-making also carries the risk of bias, transparency in automated systems could potentially improve fairness by allowing data correction, although it may also increase costs.

The feasibility of fair credit scoring in a profit-driven environment has also been explored, with researchers examining the tension between fairness and profitability. Kozodoi et al. [17] examined this issue and found a strong inverse relationship between the two factors, suggesting that reducing discrimination in credit scoring systems could be achieved at a relatively low cost for financial institutions. However, they caution that creating a fully fair system would limit profitability and increase the risk of defaults. Liu et al. [24] further explored the potential impact of fairness constraints on credit scoring, emphasizing that while fairness measures may be intended to protect specific groups, they could inadvertently harm both consumers and financial institutions over time. They argue that dynamic modeling of fairness criteria could mitigate these negative effects, providing a more balanced approach. Creager et al. [25] proposed a causal modeling framework that simulates various scenarios in profit-driven yet policy-constrained environments, offering insights into how fairness can be evaluated over the long term in both institutional and individual contexts. Another critical issue in the fair credit scoring literature is the potential bias in data, which may not be apparent through mathematical approaches alone. Lee and Floridi [27] argue for leveraging the context-dependency of data, showing that some algorithms fail to ensure fairness due to the relationship between protected attributes and other features in the dataset. While the study assumes that lenders aim to maximize loan value, the profitability of the lending business as a whole must also be considered. Kilbertus et al. [29] examined the challenges of balancing fairness and profit in credit scoring, especially when previous decisions may be biased. They proposed a system using stochastic decision rules to improve both utility and fairness in credit scoring. Additionally, Szepannek and Lubke [30] introduced a group unfairness index to measure and compare the fairness of different models, with a focus on group fairness by acceptance rate, which is particularly relevant for evaluating credit scoring systems.

## III. METHODOLOGY

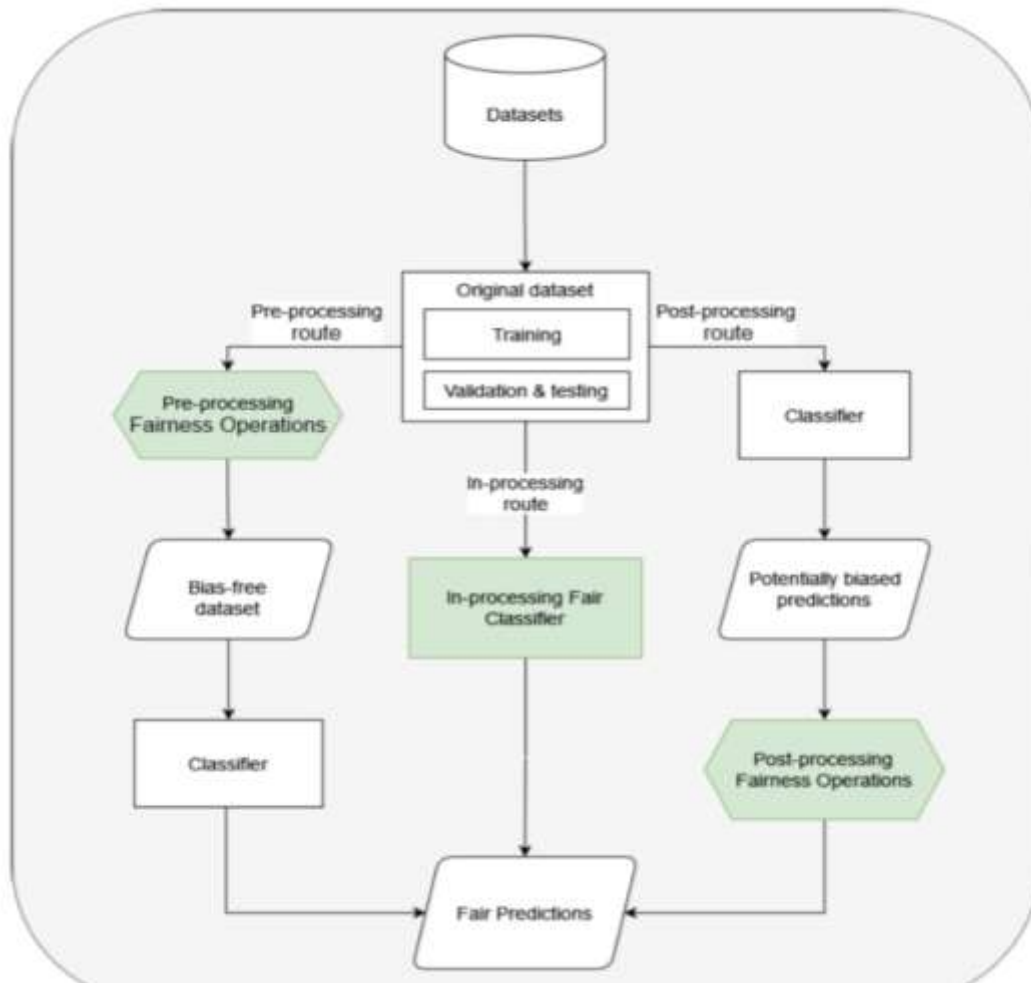
The methodology for ensuring fairness in algorithmic credit scoring involves a three-stage pipeline: bias assessment, bias mitigation, and evaluation of results. This pipeline has been widely adopted in the literature to systematically identify and address biases in machine learning models (Barocas & Selbst, 2016). Figure 1 illustrates the fairness pipeline, where algorithms are first evaluated using specific fairness metrics, then subjected to mitigation techniques, and finally assessed based on post-mitigation performance.

### 3.2 Bias Assessment Metrics

#### a. Separation Metrics

Separation metrics focus on evaluating the classification model's ability to ensure equality of opportunity across different groups. A commonly used metric is the **Average Odds Difference** (Hardt et al., 2016):

$$SP = \frac{1}{2} |(FPR_{(D = unprivileged)} - FPR_{(D = privileged)}) + (TPR_{(D = unprivileged)} - TPR_{(D = privileged)})|$$



**Figure 1:** Fair processing pipeline

Where FPR and TPR represent the false positive rate and true positive rate, respectively. A fair classification typically results in values close to zero.

#### b. Equal Opportunity Difference

This metric measures the difference in true positive rates (TPR) between protected and unprotected groups:

$$EOD = TPR_{(D = \text{unprivileged})} - TPR_{(D = \text{privileged})}$$

Values within the range (-0.1, 0.1) are generally considered fair (Kleinberg et al., 2018).

### 3.3 Independence Metrics

#### a. Statistical Parity Difference

Statistical parity assesses fairness by comparing the acceptance rates between groups:

$$SPD = Pr(\hat{Y} = 1 | D = \text{unprivileged}) - Pr(\hat{Y} = 1 | D = \text{privileged})$$

Where values near zero indicate balanced outcomes across groups (Agarwal et al., 2020).

#### b. Disparate Impact

This ratio-based metric evaluates the relative likelihood of favorable outcomes for different groups:

$$DI = \frac{Pr(\hat{Y} = 1 | D = \text{unprivileged})}{Pr(\hat{Y} = 1 | D = \text{privileged})}$$

A value between 0.8 and 1.25 is typically acceptable as fair (Zafar et al., 2017).

### 3.4 Individual Fairness Metric

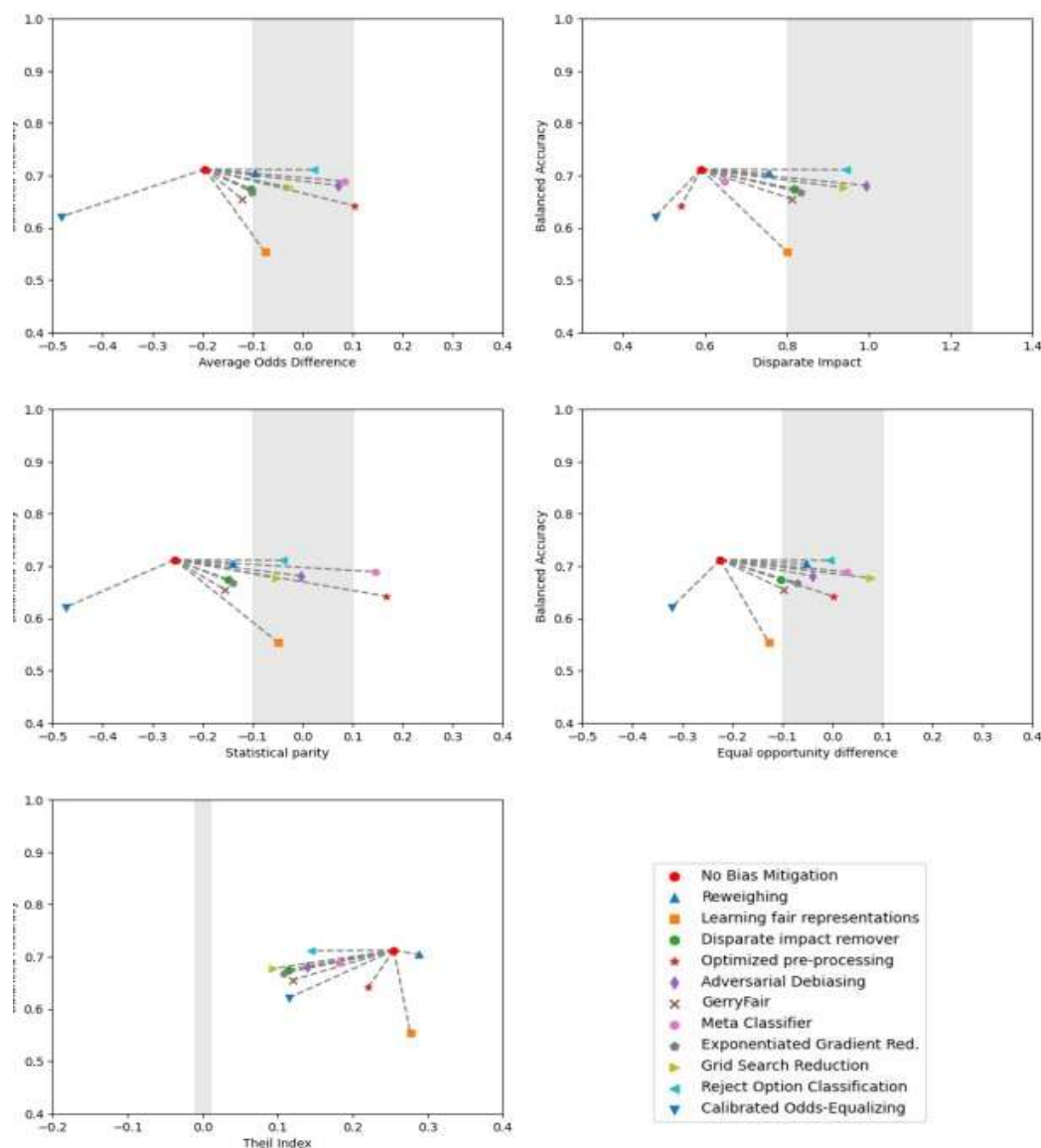
The **Theil Index**, derived from economic inequality measures, assesses both group-level and individual-level disparities (Speicher et al., 2018):

$$Theil = \frac{1}{n} \sum_{i=1}^n \frac{b_i}{\mu} \ln \frac{b_i}{\mu}, \text{ where } b_i = \hat{y}_i - y_i + 1$$

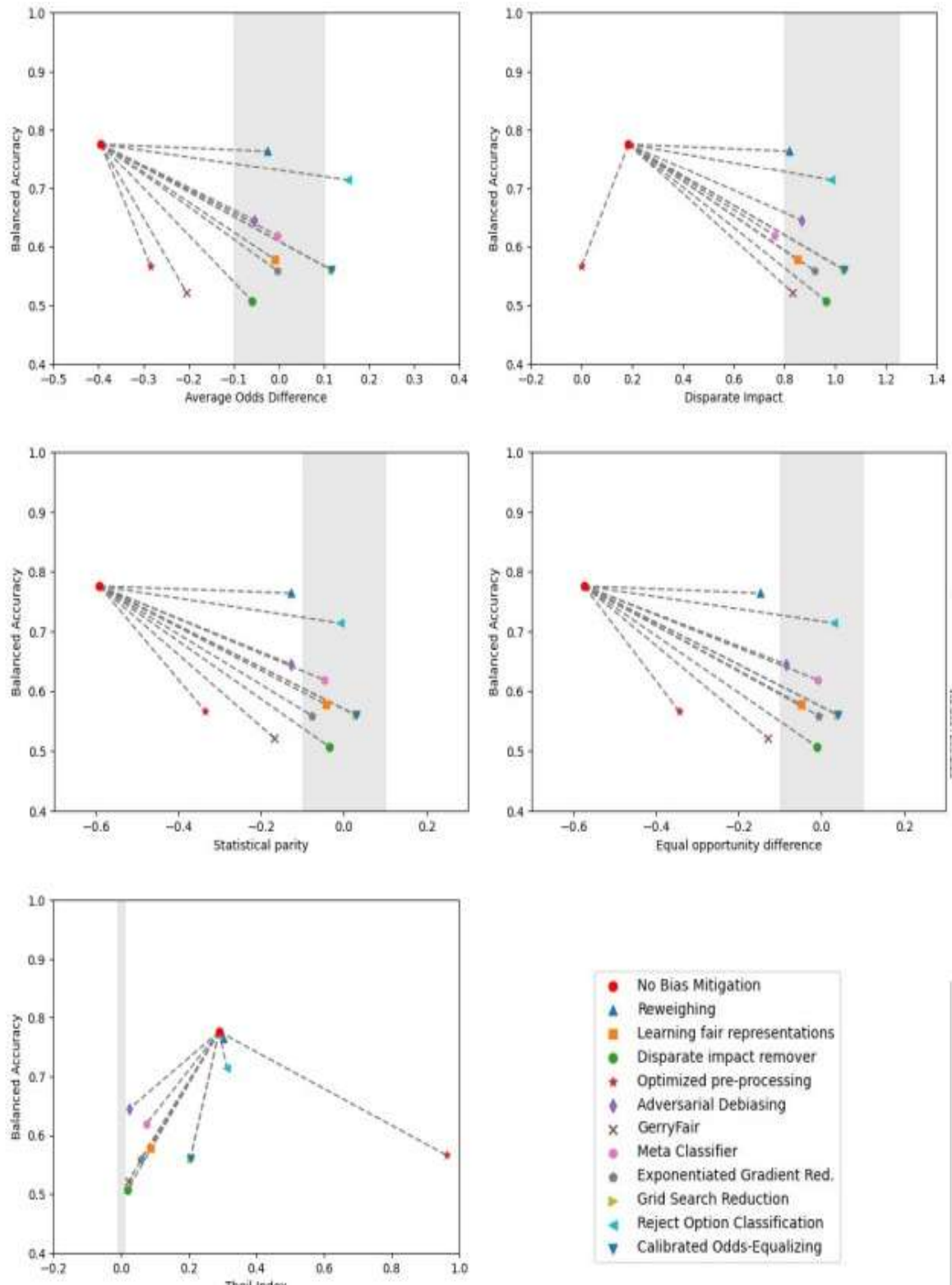
A lower Theil Index suggests a fairer model.

### 3.5 Bias Mitigation Methods

Bias mitigation methods are classified into three categories based on their position in a fair AI process (see Figure 1). The pre-processing methods address fairness issues before applying the classification method, and are therefore independent of the classification algorithm. To ensure fairness, in-processing methods have been developed within the classification method. The post-processing bias mitigation methods were placed at the end of the fair processing pipeline. They made adjustments after the model was trained considering the protected attribute restrictions. Post-processing methods, such as pre-processing methods, are independent of the classification algorithm used, permitting a classifier agnostic approach to mitigate bias. The methods presented below were chosen from among those presented in highly cited studies on this topic.



**Figure 2:** Performances bias mitigation methods considering the 5 fairness metrics on the german dataset



**Figure 3:** Performances bias mitigation methods considering the 5 fairness metrics on the loan dataset

Bias mitigation methods can be categorized into three types based on their implementation stage:

**Pre-processing Methods:** Pre-processing methods aim to mitigate bias by modifying the input data before it is used for model training. One common technique is **Reweighting**, which adjusts the weights of training samples to reduce discrimination without altering the labels (Kamiran & Calders, 2012). The algorithm assigns weights based on the ratio of expected and observed probabilities for instances with protected attributes:



$$W(X) = \frac{P_{exp}(S = X(S) \wedge Class = X(Class))}{P_{obs}(S = X(S) \wedge Class = X(Class))}$$

Another widely used pre-processing approach is **Learning Fair Representations**. This technique encodes data into a new space, obfuscating information about protected attributes (Zemel et al., 2013). The goal is to ensure that the probability of mapping to a prototype is independent of the protected group:

$$P(Z = k|x^+ \in X^+) = P(Z = k|x^- \in X^-), \forall k,$$

Where denotes the latent representation, and are instances from protected and unprotected groups, respectively. Additionally, the **Disparate Impact Remover** algorithm (Feldman et al., 2015) corrects data by reducing the disparate impact while maintaining predictability.

**In-processing Methods:** In-processing methods directly incorporate fairness constraints during model training. **Adversarial Debiasing** introduces an adversary that attempts to predict the protected attribute while the classifier tries to minimize both the classification loss and the adversary's accuracy (Zhang et al., 2018). The **GerryFair** algorithm (Kearns et al., 2018) uses a cost-sensitive learner and an auditor to iteratively detect and mitigate biases. Additionally, **Exponentiated Gradient Reduction** (Agarwal et al., 2018) minimizes error subject to fairness constraints, particularly statistical parity and equalized odds. **Post-processing Methods:** Post-processing methods apply adjustments to model predictions to achieve fairness. **Reject Option Classification** (Kamiran et al., 2012) relabels instances in uncertain regions based on their protected attributes. This method uses cost-sensitive matrices to optimize a discrimination-accuracy trade-off parameter:

$$L_{+,-}^d = L_{-,+}^f = \theta/(1 - \theta)$$

Another prominent technique is **Calibrated Equalized Odds Post-processing** (Pleiss et al., 2017), which adjusts prediction probabilities to ensure fairness while preserving accuracy. It maintains balance in false positive and false negative rates across groups using the equalized odds metric.

### 3.6 Dataset Overview and Characteristics

We used the German credit dataset, which is one of the most commonly used datasets for benchmarking in the fair AI domain. It has also been included in other research on fair AI, such as studies by [30, 43, 17]. Real-world credit scoring datasets are often scarce and present various business-specific challenges like class imbalance and profit ratio disparities. To address this, we introduced a novel consumer loan dataset for the community's reference. Table I provides a summary of the statistics for both datasets.

**Table 1:** Summary Statistics for the Datasets

| Dataset        | Instances | Cat/Num/Bin<br>Attributes | Default<br>Ratio | Protected<br>Attribute | Protected<br>Group Ratio |
|----------------|-----------|---------------------------|------------------|------------------------|--------------------------|
| German credit  | 1000      | 32/7/1                    | 30%              | Age                    | 19%                      |
| Consumer loans | 21568     | 7/7/9                     | 5.7%             | Age                    | 4.09%                    |

Both datasets contain samples of loan applications, with the target variable representing loan outcomes, classified as good (repaid loans) or bad (unrepaid loans). Independent attributes in these datasets are categorized into sociodemographic factors (e.g., age, education, and profession), customer relationship history (e.g., previous loans and banking products), and economic indicators (e.g., income and loan amount).

### 3.7 Selection of Protected Attributes

In line with previous studies [44, 17, 43], we considered the age of the applicant as a protected attribute. The age threshold of 25 years was used to define the vulnerable group (below 25 years) and the non-vulnerable group (25 years and above) [45]. When analyzing the importance of variables in relation to the target variable (Default Flag) for the consumer loan dataset, age received the highest score, with a log-odds value of **73.736**. Marital status followed with a score of **56.937**. Although previous works [3] suggest using gender as a protected attribute, we deemed this inappropriate as using attributes such as gender and race for decision-making is explicitly prohibited by legal frameworks globally, promoting fairness through unawareness.

### 3.8 Data Preprocessing and Curation

The preprocessing phase involved cleaning and transforming the datasets. For the consumer loan dataset, irrelevant attributes like ID, birthplace, and profession were dropped due to their limited predictive power and increased noise. Categorical attributes with missing values were imputed with a new class labeled "missing." For numerical attributes, missing data were replaced using median imputation to account for skewed distributions. Furthermore, one-hot encoding was applied to categorical variables to convert them into numerical representations, ensuring compatibility with various algorithms. The German credit dataset required minimal preprocessing as it was already well-curated.

### 3.9 Experimental Setup

The data for consumer loans was randomly partitioned into a training set (70%), validation set (15%), and test set (15%) using stratified sampling to maintain the target variable distribution. Due to the smaller size of the German credit dataset, it was split into a training set (70%) and a test set (30%). We checked for statistical parity differences between the protected and unprotected groups to ensure balanced conditions. Table II displays the parity differences for both datasets.

**Table 2: Statistical Parity Differences**

| Dataset        | Training | Validation | Test    |
|----------------|----------|------------|---------|
| German credit  | -0.1260  | N/A        | -0.1335 |
| Consumer loans | -0.1971  | -0.2050    | -0.1997 |

### 3.10 Evaluation Metrics

We assessed the effectiveness of bias mitigation techniques using fairness-specific metrics, classification accuracy, and profit metrics. Balanced accuracy was employed to measure model performance, as it provides a better estimate for imbalanced datasets. Equation below defines balanced accuracy as follows:

$$\begin{aligned} \text{Balanced accuracy} &= \frac{1}{2} \left( \frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \\ &= \frac{1}{2} (TPR + TNR) \end{aligned}$$

Profit calculations used the Return on Investment (ROI) metric [46] to estimate gains from correct classifications of good clients. Equation below defines ROI using factors like interest rates, loan term, and an early repayment coefficient (ERC):

$$\text{ROI} = \text{INTREST RATE} \times \text{\#YEARS} \times \text{ERC}$$

Losses from misclassifications (false positives) were factored using a loss coefficient (LC) set at **0.9**. Equation below computes the profit as follows:

$$\text{PROFIT} = \text{TPR} \times \text{ROI} - \text{FPR} \times \text{LC}$$

The same values were applied to both datasets to ensure comparability, as practiced in prior research by Kozodoi et al. [17]. Table III presents the performance metrics and their respective fairness intervals.

**Table 3: Performance Metrics**

| Metric                        | Fairness interval | Description                          |
|-------------------------------|-------------------|--------------------------------------|
| Average odds difference       | (-0.1, 0.1)       | Separation metric                    |
| Equal opportunity difference  | (-0.1, 0.1)       | Separation metric                    |
| Statistical parity difference | (-0.1, 0.1)       | Independence metric                  |
| Disparate impact              | (0.8, 1.25)       | Independence metric                  |
| Theil index                   | $\approx 0$       | Individual fairness for independence |
| Balanced accuracy             | N/A               | Classification performance metric    |
| Profit                        | N/A               | Model's profit (%)                   |

## IV. RESULT AND DISCUSSION

The results from applying fairness-aware algorithms on the German credit and consumer loan datasets are summarized in Tables IV and V. The best values for each evaluation criterion are underlined to highlight the

most effective approaches. Analysis indicates that mitigating bias often results in a loss of accuracy and profit, though certain methods successfully balanced these trade-offs.

**Table 4: Benchmark results (Consumer loan dataset)**

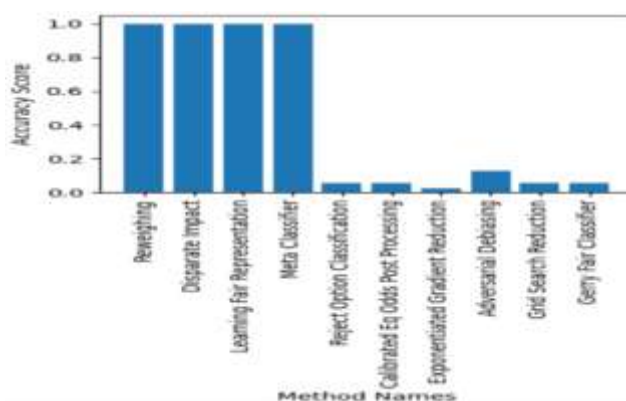
| Fairness processor            | Proc. type | DI    | SP     | AOD    | EOD    | TI    | BAcc  | P     |
|-------------------------------|------------|-------|--------|--------|--------|-------|-------|-------|
| Reweighting                   | Pre        | 0.818 | -0.127 | -0.026 | -0.148 | 0.303 | 0.764 | 0.319 |
| Learning Fair Representations | Pre        | 0.850 | -0.044 | -0.009 | -0.050 | 0.087 | 0.578 | 0.283 |
| Disparate impact remover      | Pre        | 0.964 | -0.035 | -0.059 | -0.011 | 0.019 | 0.507 | 0.270 |
| Optimized pre-processing      | Pre        | 0.000 | -0.335 | -0.284 | -0.344 | 0.962 | 0.566 | 0.278 |
| Adversarial Debiasing         | In         | 0.868 | -0.128 | -0.054 | -0.086 | 0.025 | 0.645 | 0.289 |
| GerryFair                     | In         | 0.830 | -0.169 | -0.205 | -0.131 | 0.022 | 0.522 | 0.272 |
| Meta Classifier*              | In         | 0.762 | -0.048 | -0.004 | -0.010 | 0.075 | 0.619 | 0.288 |
| Exponentiated Gradient Red.   | In         | 0.918 | -0.077 | -0.003 | -0.007 | 0.058 | 0.559 | 0.277 |
| Grid Search Reduction         | In         | 1.036 | 0.029  | 0.117  | 0.04   | 0.205 | 0.560 | 0.279 |
| Prejudice Remover*            | In         | N/A   | N/A    | N/A    | N/A    | 0.058 | 0.500 | 0.269 |
| Reject Option Classification  | Post       | 0.984 | -0.010 | 0.153  | 0.030  | 0.313 | 0.714 | 0.313 |
| Calibrated Odds-Equalizing    | Post       | 1.036 | 0.029  | 0.117  | 0.04   | 0.205 | 0.560 | 0.279 |
| No bias mitigation            | N/A        | 0.184 | -0.591 | -0.395 | -0.574 | 0.290 | 0.776 | 0.321 |

Abbreviations: DI = disparate impact; SP = statistical parity;  
AOD = average odds difference; EOD = equal opportunity difference;  
TI = Theil index; BAcc = balanced accuracy; P = profit.  
\* unstable results

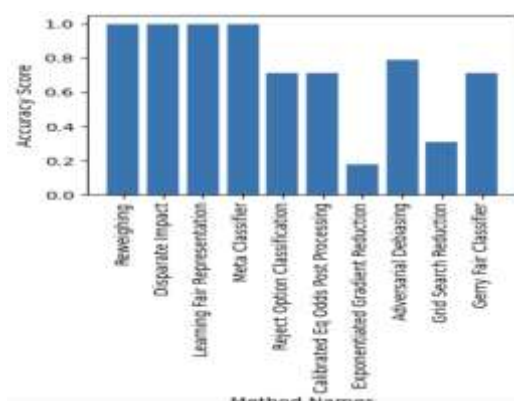
Abbreviations: DI = disparate impact; SP = statistical parity;  
AOD = average odds difference; EOD = equal opportunity difference;  
TI = Theil index; BAcc = balanced accuracy; P = profit.  
\* unstable results

In the consumer loan dataset, Learning Fair Representations, Disparate Impact Remover, and Exponentiated Gradient Reduction achieved fairness across all five metrics. Similarly, Grid Search Reduction performed well with the German credit dataset. However, some algorithms, including Prejudice Remover and Gerry Fair, exhibited inconsistent results, likely due to bias within the datasets. Notably, the German credit dataset's smaller size (1000 instances) may have amplified volatility in the Theil index values, while the consumer loan dataset's severe class imbalance (5.7% defaulted loans) hindered accurate classification. Balanced accuracy proved essential in such scenarios for evaluating performance without the bias introduced by imbalanced classes.

Figures 2 and 3 (Appendix B) provide visual comparisons of biased and de-biased values. These plots illustrate the relationship between balanced accuracy and specific fairness metrics. Grey regions represent the desired fairness range, aiding in the assessment of mitigation effectiveness. Reweighting, as a pre-processing method, consistently demonstrated robust bias mitigation with minimal accuracy loss. Despite this, a higher-than-recommended Theil index in some cases suggested potential residual unfairness at the individual level. Learning Fair Representations also showed promise, particularly in its ability to balance multiple fairness constraints. Although parameter tuning was complex due to numerous factors like fairness constraints and classification thresholds, the flexibility to choose advanced classifiers provided additional advantages. Conversely, Disparate Impact Remover optimized the disparate impact metric effectively but at a significant cost to accuracy and profit. This trade-off underscores the need to balance fairness objectives with practical performance considerations.



**Figure 4: Accuracy for consumer loan data set**



**Figure 5: Accuracy for credit score data set**



## V. CONCLUSION

This study significantly contributes to the understanding of fair AI decision-making by benchmarking 12 bias mitigation methods using five fairness metrics in the context of credit scoring. By evaluating these methods on both the traditional German credit dataset and a novel consumer loans dataset from a Romanian bank, we highlighted the complexities and trade-offs associated with implementing fairness-aware algorithms in real-world scenarios. While most methods demonstrated the ability to enhance fairness, these improvements were often accompanied by reductions in accuracy and profitability, emphasizing the inherent challenges of bias mitigation. The analysis revealed that no single method serves as a comprehensive solution for addressing all fairness concerns while maintaining satisfactory accuracy and profit. Learning Fair Representations, Disparate Impact Remover, and Exponentiated Gradient Reduction showed consistent success in improving fairness on the consumer loan dataset, while Grid Search Reduction performed notably well on the German credit dataset. Despite these achievements, the results underscore the need for practitioners to apply multiple methods, carefully assess trade-offs, and select the most appropriate algorithm based on specific cost and fairness considerations. Additionally, the study exposed practical challenges in applying bias mitigation methods to real-world data. Many existing studies primarily focus on simple accuracy metrics, which proved insufficient in our highly imbalanced datasets. The consumer loan dataset's severe class imbalance (with only 5.7% defaulted loans) further complicated the evaluation process, demonstrating the necessity of using balanced accuracy to provide a more accurate reflection of model performance. Cost-sensitive classification methods may offer a viable approach to address this issue, providing a more balanced trade-off between fairness, accuracy, and profitability. Future research should focus on developing adaptive algorithms that dynamically balance these objectives across various datasets and contexts. Experimenting with a range of classifiers beyond logistic regression could also yield more accurate results, as classifier selection plays a crucial role in determining overall performance. Furthermore, assessing the environmental impact of computationally intensive methods by calculating their energy consumption and carbon footprint will become increasingly relevant. Incorporating sustainability considerations alongside fairness metrics, accuracy, and profit will enable practitioners to make more informed and responsible decisions in deploying AI-based credit scoring systems.

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